

First Order Linear Differential Equations

The standard form of a first order linear differential equations is

$$\frac{d y}{d x} + P(x) y = Q(x)$$

Here P and Q are continuous functions of x only.

$$\frac{d y}{d x} + 2 x y = x^2$$

Linear

$$\frac{d y}{d x} + 2 x \sqrt{y} = 2 x$$

Non linear

$$\frac{d y}{d x} + 2 x y^2 = \tan x$$

Non linear

$$\frac{d y}{d x} + P(x) y = Q(x)$$

$$P(x) y dx - Q(x) dx + dy = 0$$

$$\underbrace{[P(x) y - Q(x)]}_{M} dx + \underbrace{1}_{N} . dy = 0$$

$$\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)}{N} = \frac{P(x)}{1} = P(x)$$

$$I.F. = e^{\int P(x) dx}$$

Example 1

$$\frac{dy}{dx} + 2x y = 2e^{-x^2}$$
$$\frac{dy}{dx} + P(x) y = Q(x)$$


This is a first order linear differential equation.

$$IF = e^{\int P(x) dx}$$

$$\int P(x) dx = \int 2x dx$$

$$IF = e^{x^2}$$

Multiplying with the integrating factor the equation becomes

$$e^{x^2} \frac{dy}{dx} + 2xy e^{x^2} = 2$$

Integration gives

$$d(e^{x^2} y) = 2 dx$$

$$e^{x^2} y = 2x + c$$

Example 2

$$\frac{dy}{dx} + \frac{1}{x} y = x^2$$

Example 3

$$(2x - 10y^3) \frac{dy}{dx} + y = 0$$